

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 2nd Semester B.Sc. Final Examination, 2023
CSE 1201: Fundamentals of Programming

Total Marks: 70

Time: 3 Hours

[The figures in the right margin represent the full marks, and the symbols used follow standard notation]

(Answer any 5 (Five) of the following Questions)

1.
 - a) "C language is well suited for structured programming", explain why? 4
 - b) What is a token? What are different types of tokens available in C language? 3
 - c) Evaluate the following expressions: 2+2
 - i) $22 + 3 < 6 \ \&\& \sim 5 \ || \ 22 == 7 \ \&\& \ 22 - 2 > +5$
 - ii) $a + 2 > b \ || \ \sim c \ \&\& \ a == d * a - 2 < = e$, Where $a=11$, $b=6$, $c=0$, $d=7$ and $e=5$.
 - d) Write the output of the following C program: (Take 7348 as input) 3

```
#include<stdio.h>
int main()
{
    int n, reverse=0, rem;
    printf("Enter a number: ");
    scanf("%d", &n);
    while(n!=0)
    {
        rem=n%10;
        reverse=reverse*10+rem;
        n/=10;
    }
    printf("Reversed Number: %d",reverse);
    return 0;
}
```
2.
 - a) Using bitwise operator write a program in C to check whether a number is even or odd. 3
 - b) Differentiate between the entry-controlled loop and exit-controlled loop. 3
 - c) Explain the syntax of the switch case construct. Write a menu-driven 'C' program using a switch case to take 2 integers as input and print the sum, difference, and product depending upon the choice provided by the user. 5
 - d) Calculate the output for the following code: 3

```
#include <stdio.h>
int main()
{
    int a = 100, b = 200, c = 300;
    int x = (a>b) ? ((a>c) ? a:c) : ((b>c) ? b:c);
    printf("%d",x);
}
```
3.
 - a) Define Array. Explain the declaration of various kinds of Arrays with example. 5
 - b) What will be the output from the following C program? 3

```
#include <stdio.h>
void printArray(int arr[])
{
    printf("Size of Array in Functions: %d\n", sizeof(arr));
    printf("Array Elements: ");
    for (int i = 0; i < 5; i++)
    {
        printf("%d ",arr[i]);
    }
}
int main()
{
    int arr[5] = { 10, 20, 30, 40, 50 };
    printf("Size of Array in main(): %d\n", sizeof(arr));
    printArray(arr);
    return 0;
}
```

- c) Write a C program to find the largest value in an array of elements by using function. 4
d) Mention the rules to pass an array to a function. 2
4. a) Write down the syntax and draw the flowchart for the following looping statements: 2 x 3
i) **while** statement ii) **do-while** statement iii) **for** statement.
b) Re-write the following code with "**do-while**" loops only. 4
- ```

int i, j;
for(i=1; i<=5; i++)
{
 for(j=1; j<=i; j++)
 {
 printf("%d", i);
 printf("\n");
 }
}

```
- c) Define the **switch** statement and **goto** statement with examples. 4
5. a) Write the basic structure of **calloc( )**, **malloc( )** function. Which is more preferable to use and why? 3  
b) Explain null pointer and void pointer in C with example. 3  
c) Write the output of the following code: 2
- ```

int array [] = {6, 7, 8, 9, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
int *p = array +5;
printf( "%d\n", p[3] );

```
- d) Compare between **break** and **continue** with proper examples. Write a C program to print all even numbers from 120 to 0 except 14 and 98. 6
6. a) Write a C program to take details (id, date of birth, position) of n employees in a structure array and show all the details. Use a structure named **Employee**. 4
b) Write a program in C using pointer to search the smallest element in an array of numbers. 4
c) What is the difference between static memory allocation and dynamic memory allocation? 3
d) Write a function that displays a solid square of asterisk whose side is specified as an integer parameter side. For example , if the side is 4, the function displays: 3
- ```


```
7. a) Write a C program to define Function-like-Macro to find the maximum of two numbers. 4  
b) Write a single statement to accomplish each of the following. Assume that each of these statements applies to the same program. 4
- Write a statement that opens file "previousAC.dat" for reading and assigns the returned file pointer to **ofPtr**.
  - Write a statement that opens file "transferAC.dat" for reading and assigns the return file pointer to **tfPtr**.
  - Write a statement that opens file "newAC.dat" for writing and assigns the returned file pointer to **nfPtr**.
  - Write a statement that reads a record from the file " previousAC.dat". The record consists of integer **accountNum**, string **name**, and floating point **currentBalance**. [Note that name consists of only alphabets]
- c) Write the output of the following code: 3
- ```

#include <stdio.h>
#define MAC (10 + (80 / 2) * 2)
#define CAL (5 + (6 * 2) - 2)
int main()
{
    int A = 45;
    A = MAC + CAL;
    printf("%d", A);
    return 0;
}

```
- d) Why do we use **free()** and **realloc()** in dynamic memory allocation in C? Explain. 3

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1st Year 2nd Semester B.Sc. Final Examination, 2023
CSE 1202: Digital Logic Design

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Given the two binary numbers $X=1010100$ and $Y=1000011$, now perform $X-Y$ using 2's and 1's complement. 4
- b) Convert the following binary numbers to the indicated bases 6
 - i. 10110001101.111100000110 to octal
 - ii. 0011000001101.1101 to hexadecimal
 - iii. 110111011.001010100 to octal
- c) Explain the concept of a universal gate with proper examples. 4
2. a) Simplify the following function with Karnaugh map- 4

$$F(w, x, y, z) = \sum m(2, 4, 10, 12, 14) + \sum d(0, 1, 5, 8)$$
- b) Define prime and Essential prime implicants with a proper example. 2
- c) Determine the prime and essential prime implicants of the function $F(w, x, y, z) = \sum m(1, 4, 6, 7, 8, 9, 10, 11, 15)$ using Quine–McCluskey method. 6
- d) Design AND, Ex-OR gate by using NAND gate. 2
3. a) Define Minterms, Maxterms, Sum of products and Product of sum with example. 2
- b) Define duality principle. Simplify the given Boolean function 4
 - i) $AB + ABC + \bar{A}B + A\bar{B}C$
 - ii) $A + \bar{A}B + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}D$
 - iii) $\bar{A}C + \bar{A}B + A\bar{B}C + BC$
- c) Write down the difference between canonical form and standard form. Convert the following to other canonical form 5

$$F(A, B, C) = \sum(1, 4, 5, 6, 7)$$

$$F(A, B, C, D) = \sum(0, 2, 6, 11, 13, 14)$$
- d) Express the Boolean function $F = XY + \bar{X}Z$ in a product of maxterm. 3
4. a) Construct a 4X16 decoder with 3X8 decoder. 5
- b) Use a PROM to implement the following functions: 5

$$A(x, y, z) = \sum m(1, 2, 4, 6)$$

$$B(x, y, z) = \sum m(0, 1, 6, 7)$$

$$C(x, y, z) = \sum m(2, 6)$$

$$D(x, y, z) = \sum m(1, 2, 3, 5, 7)$$
- c) Compare Asynchronous counter and Synchronous counter. Design a 3 bits ripple counter. 4
5. a) Define combinational logic circuit and classify it. 3
- b) Design a 4-bit subtractor circuit using full adder circuit. 3
- c) What is comparator? Implement a 2-bit comparator. 4
- d) Implement the full adder circuit with a decoder and two OR gates. 4
6. a) Define multiplexer and demultiplexer. Implement the 8 to 1 line multiplexer circuit. 5
- b) A combinational circuit is defined by the function $F_1(A, B, C) = \sum(3, 5, 6, 7)$ and $F_2(A, B, C) = \sum(0, 2, 4, 7)$ Implement the circuit with PLA having 3 input, four products terms and two outputs. 5
- c) Define sequential circuit and described the operation of JK flip - flop with figure. 4
7. a) Differentiate between Static RAM and Dynamic RAM. 4
- b) Define Shift Register. Design a synchronous counter that has the following counting sequence: 6

$$000, 010, 101, 110$$
- c) Show the logical construction of a 4X4 RAM. 4

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MATH – 1204: Methods of Integration, Differential Equations and Series

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Evaluate the following integrals: 6
 i) $\int (\sin^{-1} x)^2 dx$ ii) $\int \sin^4 x \cos^3 x dx$
 b) Solve: $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$. 5
 c) Evaluate the improper integral $\int_1^{\infty} \frac{1}{\sqrt{x}} dx$. 3

2. a) Solve: $(x^2 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0$. 5
 b) Find the differential equation of the curves $y = e^x(A \cos x + B \sin x)$. State order and degree of the derived equation. 5
 c) Expand $2x^3 + 7x^2 + x - 6$ in powers of $(x - 2)$ by Taylor's Theorem. 4

3. a) Expand $\log(1+x)$ in power of x by Maclaurin theorem. 4
 b) What is Taylor's infinite series? Find the first three terms of the Taylor series for the function $\ln x$ at $x = 1$ 5
 c) Show that arc length of the parabola $r(1 + \cos\theta) = 2$ from $\theta = 0$ to $\theta = \frac{\pi}{2}$ is 5
 $\sqrt{2} + \ln(1 + \sqrt{2})$

4. a) Solve the following differential equation $\frac{dy}{dx} + \frac{1}{x}y = x\sqrt{y}$. 6
 b) In a certain bacteria culture the rate of increase in the number of bacteria is proportional to the number present. If the number triples in 5 hours, how many will present in 10 hours? 6
 c) Define Integrating factor. 2

5. a) Using Taylor's series method solve: $\frac{dy}{dx} = x^2 - y$ 7
 With $y(0) = 1$, numerically up to $x = 0.2$ with $h = 0.1$
 b) Find $y(2.2)$ using Euler's method from the equation 7
 $\frac{dy}{dx} = -xy^2$; $y(2) = 1$ with $h = 0.05$

6. a) If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$ 2
 b) Test the convergence of the following series (any two): 6

$$(i) \sum_{n=1}^{\infty} \frac{n^2}{5n^3 + 4} \quad (ii) \sum_{n=1}^{\infty} \frac{1}{n^2 + 1} \quad (iii) \sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$$

 c) Find a rule of general solution of differential equation $\frac{dy}{dx} + py = Q$, where P and Q are 6
 only functions of x or constants. Hence find the solution of $(1+x^2)\frac{dy}{dx} + y = \tan^{-1} x$

7. a) Define order and degree of the differential equation. 2
- b) State and prove that the necessary conditions for exactness of the differential equation $Mdx + Ndy = 0$ 3
- c) Solve the following differential equations (any three): 9

(i) $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y$

(ii) $x\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy = 0$

(iii) $y(1+xy)dx + x(1-xy)dy = 0$

(iv) $\frac{dy}{dx} + \frac{y^2 + y + 1}{x^2 + x + 1} = 0$

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PHY 1203: Physics

Total Marks: 70

Time: 3 Hours

[The figures in the right margin represent the full marks, and the symbols used follow standard notation.]

(Answer any 5 (Five) of the following Questions)

1. a) What is an irreversible process? Derive the expression for the work done during an isothermal expansion. 5
 b) What is entropy, and how can you demonstrate that it remains constant during a reversible process? 2+4
 c) A mass of gas at 80^o C and 50 atmospheric pressure is adiabatically expanded to ten times its initial volume. If the ratio of specific heat is 1.4, calculate the resulting pressure and temperature. 3
2. a) What are the differences between polarized light and unpolarized light? 4
 b) From Newton's rings prove that the radius of the mth dark ring is proportional to under root of wavelength. 6
 c) Define refractive index. In the experiment of Newton's rings, diameter of the eighth ring is 0.4 cm and of the third ring is 0.2 cm. Radius of curvature of the lens is 101 cm. Calculate the wavelength of the light used. 4
3. a) Define: i) Joule's equivalent ii) thermodynamic function iii) thermodynamic system and iv) internal energy 4
 b) What is entropy? Show that entropy increases in an irreversible process whereas remains constant in a reversible process. 6
 c) A motor tyre is pumped to a pressure of 2 atmospheres at 15^oC, when it suddenly bursts. Calculate the resulting drop in temperature, given $\gamma = 1.4$. 4
4. a) Prove that the expression for the fringe width in a double-slit experiment is $\frac{\lambda D}{a}$. 5
 b) What is a Fresnel Biprism, and how does it work? What are the key differences between Fresnel diffraction and Fraunhofer diffraction? 2+4
 c) Light from a sodium vapor lamp ($\lambda = 589$ nm) forms an interference pattern on a screen 0.8 m from a pair of slits. The bright fringes in the pattern are 0.35 cm apart. What is the slit separation? 3
5. a) Explain the Band Theory of Solids and illustrate it with a diagram? What are the main differences between n-type and p-type semiconductors? 3+3
 b) How can you determine the inter-planar spacing in a crystal lattice given the wavelength of incident X-rays and the diffraction angle? Derive the equation for this. 5
 c) Sodium crystallizes in a body-centered cubic (BCC) structure. Given its density of 9.6×10^2 kg/m³ and atomic weight of 23, calculate the lattice constant of sodium. 3
6. a) Define Miller indices. Also write down some important features of Miller indices. 4
 b) What do you mean by diffraction of X-rays by crystal planes? Derive Bragg's law of diffraction. 2+5
 c) A source of sound has a frequency of 512 hz and an amplitude of 0.25 cm. What is the flow of energy across a square cm per second, if the velocity of sound in air is 340 m/s and the density of air is 0.00129g/cm³? 3
7. a) Prove that the differential equation governing two-body oscillations is identical to the differential equation of simple harmonic motion (SHM) for a single oscillating mass. 6
 b) What is meant by the degree of freedom in thermodynamics? Prove that $C_p - C_v = R$ 1+4
 c) A body describing SHM has a maximum acceleration of 8π m/s² and a maximum speed of 1.6 m/s. Find the period T and amplitude A. 3

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ENG 1205: Developing English Language Skills

Total Marks: 70

Time: 2.5 Hours

Part A (Comprehension: 15 marks)

1. Read the following passage and answer the questions given below:

Philadelphia is a city known for many things. It is where the Declaration of Independence was signed in 1776, and it was also the first capital of the United States. But one fact about Philadelphia is not so well-known: it is home to nearly 3,000 murals painted on the sides of homes and buildings around the city. It is said that Philadelphia has more murals than any other city in the world, with the exception of Rome. How did this come to be?

More than 20 years ago, a New Jersey artist named Jane Golden started a program pairing troubled youth with artists to paint murals on a few buildings around the city. From this small project, something magical happened. The young people involved helped to create magnificent pieces of art, but there were other, perhaps more important, benefits. The young people learned to collaborate and get along with many different kinds of people during the various steps required to paint and design a mural. They learned to be responsible, because they needed to follow a schedule to make sure the murals were completed. They also learned to take pride in their community. It is hard for any resident to see the spectacular designs and not feel proud to be a part of Philadelphia.

Take a walk around some of the poorest neighborhoods in Philadelphia—neighborhoods full of broken windows and littered front steps—and you will find beautiful works of art on the sides and fronts of buildings. But murals are not only found in poor neighborhoods. They are found in affluent ones as well. Special buses take tourists to different parts of the city to see the various murals, which range from huge portraits of historical heroes, to cityscapes, to scenes depicting the diverse ethnic groups that call Philadelphia home.

As a result of its success, the mural program created by Jane Golden has now become the nation's largest public art program and a model for other cities throughout the country seeking to help troubled youth.

- a) Choose the correct answer: 5
- i. This passage focuses mostly upon
 - a) an art program designed to help troubled youth
 - b) the tourists who come to Philadelphia
 - c) the many reasons why Philadelphia is a unique city
 - d) how Jane Golden came up with the idea to start a mural program
 - ii. In paragraph 1, the author states, "It is said that Philadelphia has more murals than any other city in the world, with the exception of Rome." Using this information, we can conclude that
 - a) Rome has fewer murals than Philadelphia
 - b) Philadelphia has fewer murals than Rome
 - c) Rome has the most beautiful murals in the world
 - d) Rome and Philadelphia are the only cities with murals in the world
 - iii. According to the passage, the mural program helps troubled youth by teaching them
 - a) to be more responsible
 - b) how to collaborate with others
 - c) to take pride in their community
 - d) all of a) , b) and c)
 - iv. As used in paragraph 3, which is the best definition for **affluent**?
 - a) popular
 - b) clean
 - c) well-known
 - d) wealthy
 - v. What types of murals can be found in Philadelphia?
 - a) portraits of historical heroes
 - b) cityscapes
 - c) scenes depicting diverse ethnic groups
 - d) all of a) , b) and c)
- b) Answer the following questions: 5
- i. Who was the artist that started the mural program in Philadelphia?
 - ii. How many murals are estimated to be in Philadelphia?
 - iii. What are some of the benefits that the young people gained from participating in the mural program?

- iv. Where can tourists go to see the various murals in the city?
 - v. What has the mural program in Philadelphia become known as in the nation?
- c) Write a summary of the passage. 5

Part B (Grammar: 25 marks)

2. Fill in the gaps with the correct form of verbs:

Air (a) _____ (be) an important element of the environment. But it is constantly be (b) _____ (pollute) in many ways. We make fire to cook food and (c) _____ (do) _____ my works. This fire emits smoke which (d) _____ (contaminate) air. Vehicles pollute a great extent by (e) _____ (burn) fuel. Air pollution also (f) _____ (happen) when coal _____ oil (g) _____ (burn). Mills and factories can be (h) _____ (hold) responsible for this pollution. It is high time we (i) _____ (check) air pollution. Otherwise, we (j) _____ (suffer) a lot.

3. Change the following into indirect speech: 5

The teacher said to the students, "Have you prepared your lesson today?" "No, sir," replied the students. The teacher said, "Why do you not prepare your lesson?" "I was suffering from a headache yesterday. I shall do my work tomorrow," replied the student.

4. Fill in the blanks with appropriate preposition: 5

- a) The shop doesn't have the toys I was looking _____.
- b) The teacher divided the sweets _____ all the children.
- c) The dog is grateful _____ its owner.
- d) My brother's anniversary is _____ 5th November.
- e) The boy _____ the store is quite young.
- f) Rupa is fond _____ muffins.
- g) The cat jumped _____ the sofa.
- h) Humpty Dumpty sat _____ a wall.
- i) The Sun will not rise _____ 6 o'clock.
- j) Priya's house is _____ mine.

5. Transform any five (05) of the following sentences as directed. 5

- a) If you don't leave now, you will get caught in the rain. (simple)
- b) Though we were not sure if we could finish it, we volunteered to help them. (compound)
- c) I looked for Raju everywhere but I could not find him. (complex)
- d) To finish your project in time, you should start now. (complex)
- e) Maria was a nurse and so her job was to take care of her patients. (simple)
- f) My cousins and I went for a movie yesterday as we were bored. (compound)

6. Fill in the gaps with appropriate articles (a/an/the). Put a cross (x) if it requires. 5

- a) Always speak _____ truth and never tell _____ lie.
- b) Their car does 150 miles _____ hour.
- c) Do you still live in _____ Bristol?
- d) I have _____ one taka note.
- e) She is _____ BA but her husband is _____ MA.
- f) What do you usually have for _____ breakfast?
- g) Ben has _____ terrible headache.
- h) After this tour you have _____ whole afternoon free to explore the city.

Part C: Writing (30 marks)

7. Write an e-mail to your friend congratulating him on his achievement of a fully funded scholarship for higher study abroad. 7
8. Write a complete CV for the post of a Computer Engineer in a Company. 8
9. Write a report on "Food Adulteration" or "Cutting down Trees is a threat to our existence" 7
10. Write a paragraph on "The Role of Students in the Social Reconstruction" or "Bad Effects of Using Poly bags on Environment". 8